

Towards Bias-correction in the Arctic with Deep Learning:

An Ablation Study of Sparsity for Fine-Tuning Foundation Models
in Observation-Limited Regions

1 Motivation

How much sparsity is too much sparsity?

ERA5 exhibits known biases and limitations, particularly in the sparsely observed Arctic, where near-surface temperatures (T2M) can show a warm bias of several degrees - reaching up to 5 K under cold, clear winter conditions. [1][2]

Previous studies applying ML to interpolate sparse Arctic observations have demonstrated improved performance compared to ERA5. [3][4]

In this work, we investigate how varying levels of data sparsity during training affect the pretrained transformer model AtmoRep. Our goal is to assess the potential of leveraging foundation models in data-sparse regions like the Arctic for downstream applications, such as bias correction. [5]

2 Model and Data

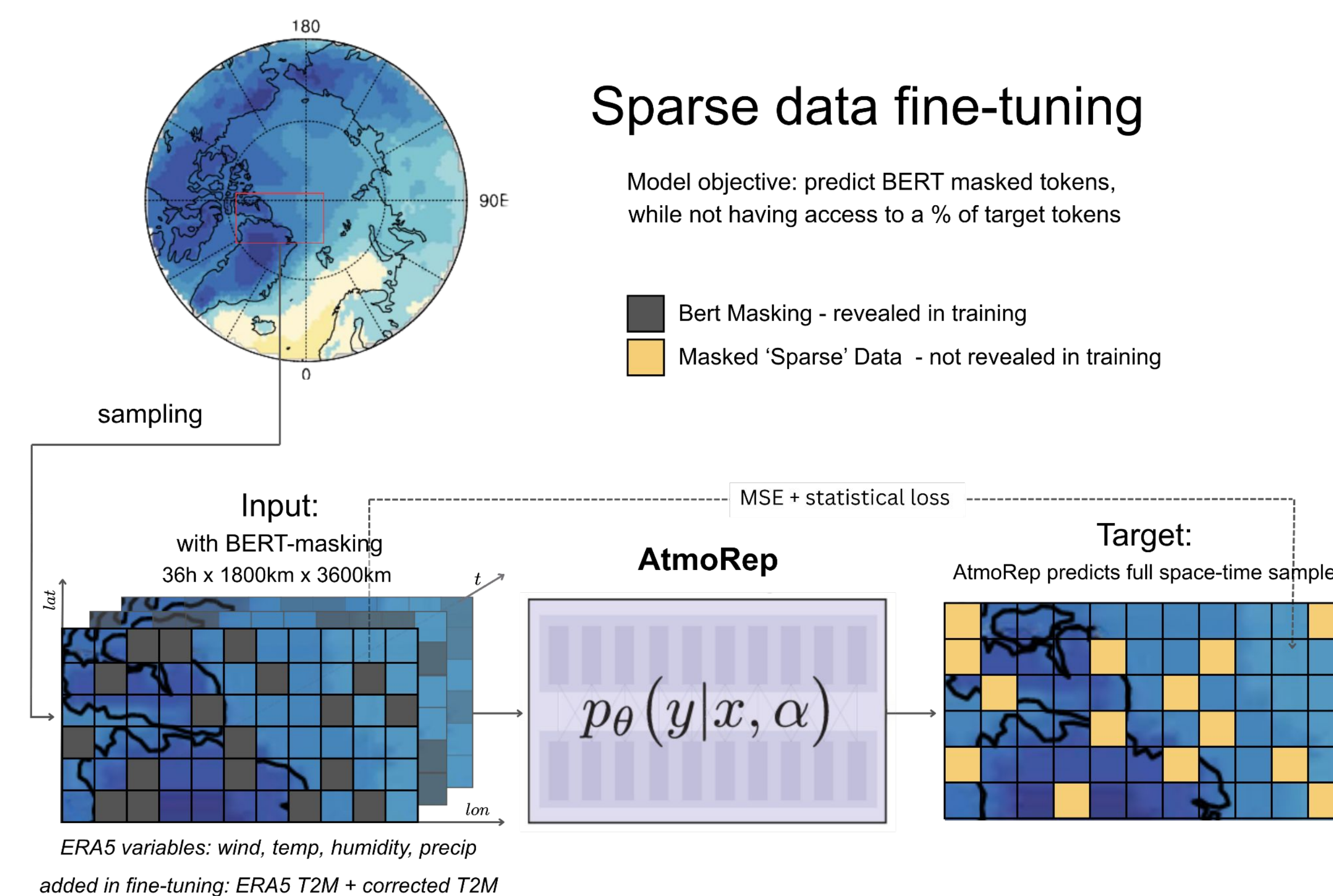


Figure adapted from Lessig et al. (2023b) [5] and Ma et al. (2023) [4]

Model: AtmoRep, a pretrained foundation model for atmospheric dynamics trained on ERA5 temp, humidity, velocity-u, -v, -z and precip from 1979-2021. [5]

Data: we apply sparsity to the bias-corrected Arctic T2M dataset presented by Hossain et al. (2025).¹ This data is as yet unseen by AtmoRep.

¹ Hossain et al.'s dataset shows a ~38% improvement of daily mean values of 2m temp in the Arctic. [3]

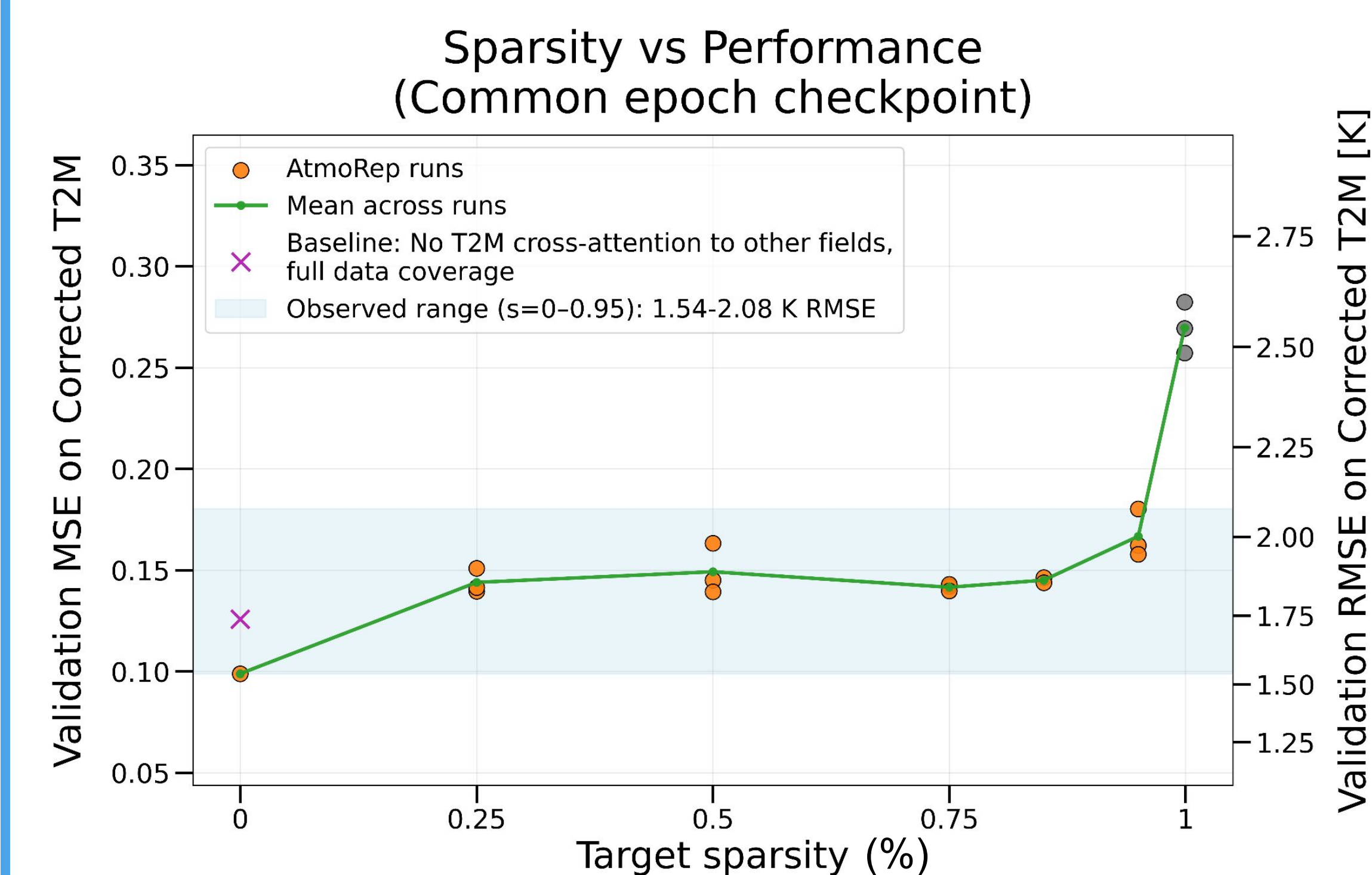
The One-Liner

We present evidence showing the potential of using sparse data for fine-tuning foundation models for downstream tasks such as ERA5 bias correction in areas with limited in situ observations.

3 Methods

1. Fine-tune the AtmoRep model on ERA5 T2M from 2010-2020 → introduce T2M to the model in the first place.
2. Apply a few-shot learning approach with bias-corrected T2M, using BERT-style masking while systematically increasing sparsity in the target space → emulating realistic data-scarce conditions, such as those from expeditions, drifting buoys, and other temporary meteorological stations.
3. Evaluate the effect on prediction performance for validation year 2021.

4 Results



Within 60 epochs AtmoRep achieves an RMSE of ~1.5-2 K across sparsity levels - already lower than typical ERA5 warm biases.

Relatively constant MSE performance from 0.25% - 0.95% target sparsity. At 100% sparsity, an expected sharp drop in performance.

5 Discussion

While these results represent exploratory research, they indicate the robust performance of the foundation model AtmoRep when fine-tuned using sparse data.

Baselines:

- The climatology baseline yields an MSE of 1 (corresponding to an RMSE of ~4.9 K).
- Removing T2M data entirely results in an expected drop in performance (dashed black line in plot above, grey points in plot left, section 4).
- Disabling cross-attention produces a smaller but still noticeable degradation even when presented with all available data (purple cross left, section 4).

Potential mechanisms:

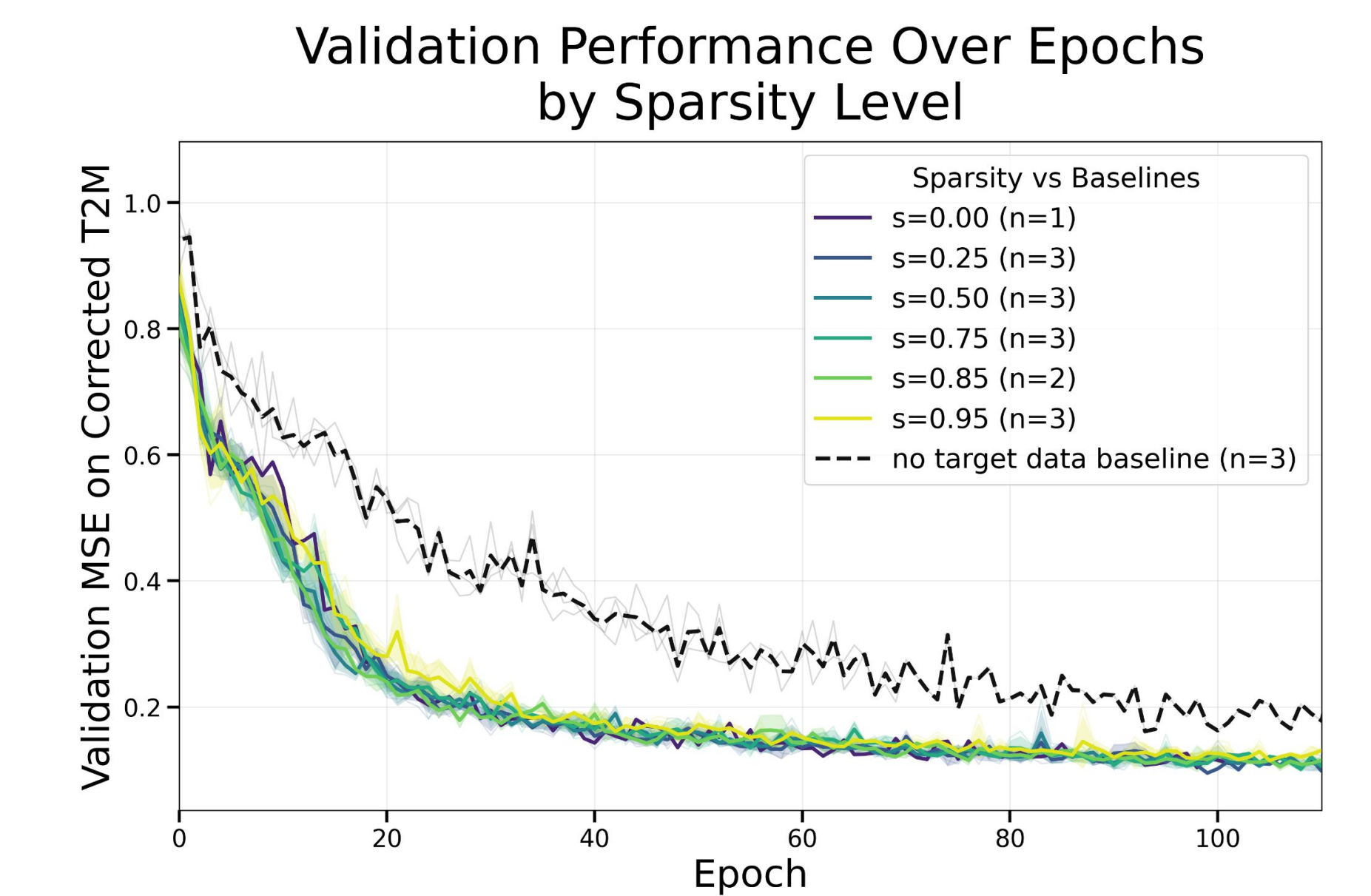
- Contextual information captured through inter-field cross-attention.
- Pretraining on seven ERA5 variables, enabling richer representations.

Potential future directions:

Mechanistic analysis: Perform more comprehensive validation, e.g. extend cross-attention removal.

Sparsity representation: Better emulate realistic Arctic sparsity using masked spatial patches instead of scattered tokens.

Forecasting: Fine-tune AtmoRep for forecasting on bias-corrected T2M data to assess ERA5 bias correction.



References:

- [1] Tian, T., Yang, S., Hoyer, J. L., Nielsen-Englyst, P., & Singha, S. (2024). Cooler Arctic surface temperatures simulated by climate models are closer to satellite-based data than the ERA5 reanalysis. *Communications Earth & Environment*, 5(1). <https://doi.org/10.1038/s43247-024-01276-z>
- [2] Herrmannsdörfer, L., Müller, M., Shupe, M. D., & Rostovsky, P. (2023). Surface temperature comparison of the Arctic winter MOSAiC observations, ERA5 reanalysis, and MODIS satellite retrieval. *Elementa Science of the Anthropocene*, 11(1). <https://doi.org/10.1525/elementa.2022.00085>
- [3] Hossain, A., Keil, P., Grover, H., Brooks, I., Cox, C. J., Gallagher, M. R., Granskog, M. A., Guy, H., Hudson, S. R., Ola, P., Persson, G., Shupe, M. D., Tjernström, M., Vüllers, J., Walden, V. P., & Pithan, F. (2025). Machine Learning Eliminates Reanalysis Warm Bias and Reveals Weaker Winter Surface Cooling over Arctic Sea Ice. *ESS Open Archive*. <https://doi.org/10.22541/essoar.176659533.30384251/v1>
- [4] Ma, Z., Huang, J., Zhang, X., Luo, Y., Ding, M., Wen, J., Jin, W., Qiao, C., & Yin, Y. (2023). Newly reconstructed Arctic surface air temperatures for 1979–2021 with deep learning method. *Scientific Data*, 10(1). <https://doi.org/10.1038/s41597-023-02059-5>
- [5] Lessig, C., Luise, I., Gong, B., Langguth, M., Stadler, S., & Schultz, M. (2023, August 25). *AtmoRep: A stochastic model of atmosphere dynamics using large scale representation learning*. arXiv.org. <https://arxiv.org/abs/2308.13280>